

Deep Learning in Video Streaming Services*

Arseniy Malyuk

Slovak University of Technology in Bratislava

Faculty of Informatics and Information Technologies

`xmalyuk@stuba.sk`

December 2024

Abstract

The purpose of this article is to explore how video streaming services such as Netflix, Amazon Prime Video, Disney+, etc. implement deep learning for personalized recommendations. The main area of focus here will be evaluation of deep learning approaches for analyzing large quantities of user data such as activity logs and preferences. We will explore how such algorithms are used for generation of accurate recommendations, increasing user satisfaction or even improving viewing rates. We will also consider using NLP and video content analysis methods for recommendation improvement. In particular, we would like to know how various streaming services operate and implement deep learning for assessing users' interaction activities with the content such as views, ratings, comments, etc. This will allow us to see the personalized recommendations that create based on users' interests and the information they already have. Towards this, we will look at some practical aspects of the deep learning application in recommender systems and evaluate the performance. This will guide us on how best such solutions can be put to use in practice and the possible limitations that might be encountered. At the end, we will discuss how improvement in these methods affects the quality and effectiveness of the streaming services. The main source of inspiration for this topic was the article "An overview of video recommender systems: state-of-the-art and research issues" [1] by Sebastian Lubos, Alexander Felfernig, and Markus Tautschnig. This article provides a comprehensive overview of current algorithms and methods in the field of video recommendation.

*Semester project in the subject Engineering Methods, academic year 2024/2025, supervisor: Pavol Baťalík

1 Introduction

Each year, television is becoming less and less popular and the initiative of content consumption is shifting to video streaming services such as Netflix, Amazon Prime Video, Disney+, etc., where you do not need to switch between channels to find what you like, but simply go to the "Recommended" section. But how exactly do recommendations work and how can they "predict" user preferences? This seems to be a magical ability, but it is actually the result of sophisticated algorithms and deep learning models that are able to analyse and learn from massive amounts of data. It is thanks to deep learning that these systems are able to accurately predict what users would like and provide unique content for each of them.

Deep learning opens up opportunities to categorise users based on their previous activity (likes, comments, shares, etc.). By detecting certain patterns, machine learning models can predict the type of content that a user is likely to view in the future.

However, the history of user activity is not the only thing that forms the basis for final recommendations. Natural Language Processing (NLP) and Video Content Analysis (VCA) are other important tools for creating personalised recommendations. NLP is used specifically to analyse the text itself, which can be found in comments, reviews, search terms, etc. This is done to help the system better understand emotions and context. On the other hand, VCA includes the study of genres, themes, and even visual elements, which helps to better understand user preferences.

Therefore, recommendation systems based on deep learning are an irreplaceable tool when it comes to analysing and predicting user preferences. However, the principle of deep learning in video streaming services is too extensive to be described in just a few sentences. For this reason, in next sections we will take a closer look at how deep learning is used in video streaming services.

2 Types of Recommendation Systems

Streaming platforms employ various types of recommendation systems, each designed to capture different aspects of user preferences and deliver personalized content. The major categories of these systems include content-based systems, collaborative filtering, hybrid approaches and group recommendations. These methods are tailored to optimize user engagement by leveraging distinct data sources and algorithms. The branching of these types of recommender systems is displayed in the Figure 1.

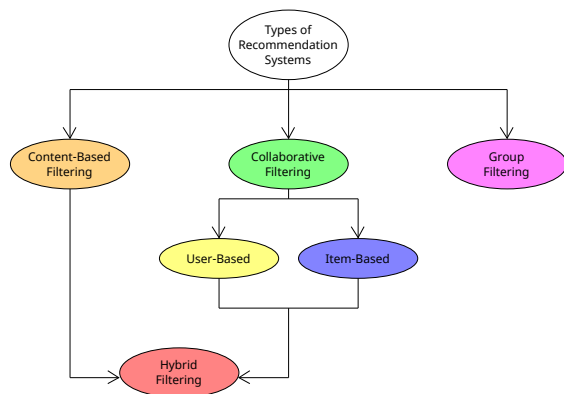


Figure 1: Branching of RS Types

2.1 Content-Based Recommendation

In content-based systems, recommendations are generated by analyzing features associated with the content itself, such as genre, cast, theme, etc. For example, if a viewer often watches historical dramas, the system will recommend similar content. These systems create user profiles based on the metadata of previously watched shows or movies, so suggestions are based on specific content attributes rather than other users' choices [1, 2].

2.2 Collaborative Filtering

Collaborative filtering suggests content, based on what similar users have watched. There are two key approaches here:

- **User-Based Collaborative Filtering:** This method recommends content to a user, based on the preferences of others with similar viewing patterns. For instance, if User A and User B share a taste in thrillers, User A's preferred shows might be suggested to User B [1, 3].
- **Item-Based Collaborative Filtering:** Instead of focusing on user similarity, item-based filtering finds similarities among shows or movies themselves. If a particular group of users has engaged with both "Game of Thrones" and "The Witcher," item-based filtering might recommend one to someone who watched the other [1, 3].

Collaborative filtering, which leverages the preferences of a large user base, is highly effective in personalizing recommendations. Netflix, for instance, has successfully implemented this method to create precise recommendations for its subscribers [4, 5].

2.3 Hybrid Recommendation Systems

Hybrid recommendation systems combine aspects of content-based and collaborative filtering to increase accuracy and relevance. By drawing on both individual and broader user base behaviors, hybrid systems address potential shortcomings of each method. Netflix's model, which uses both approaches, is a strong example of how hybrid systems can effectively balance variety and specificity in recommendations [1, 3].

2.4 Group Recommendation Systems

Group recommendations are tailored for users who watch content together. These systems aggregate individual preferences to recommend content that appeals to multiple people in a shared viewing session. For example, family or group of friends can receive suggestions based on everyone's preferences, creating a viewing experience that satisfies the group as a whole [1].

3 Implementation of Deep Learning

3.1 Neural Networks and Algorithms

Deep learning models, particularly Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) are instrumental in powering recommendation systems used by streaming services.

- **CNNs** are especially effective at processing visual and video data, enabling the extraction of relevant features from images and scenes. This capability is vital for understanding the visual elements of content, which can inform more accurate recommendations [6, 7]. The functionality of CNNs is based on a structured hierarchy of layers. The Convolutional layer applies filters to the input image, identifying and extracting essential features such as edges or textures. Following this, the Pooling layer reduces the spatial dimensions of the data, optimizing computational efficiency while retaining critical information. Finally, the Dense layer aggregates the extracted features to generate the final prediction.

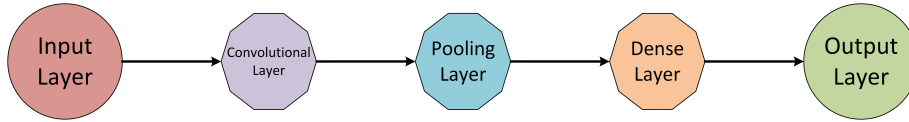


Figure 2: Simple CNN Architecture

- **RNNs** are adept at handling sequential data, such as the patterns in user viewing habits over time. By analyzing how a viewer's preferences evolve, RNNs contribute to the creation of dynamic user profiles that reflect shifting tastes [7]. In an RNN, the key component is the Recurrent Hidden layer. It processes the input at each time step while maintaining a hidden state that carries information from earlier steps. This hidden state is updated with each new input, effectively creating a memory that reflects the sequence's context.

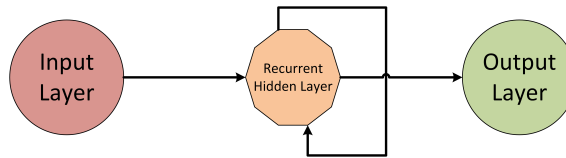


Figure 3: Simple RNN Architecture

3.2 Netflix's Recommendation System

Netflix's recommendation system is one of the most advanced in the industry, blending various deep learning techniques, such as variational autoencoders and multi-head attention mechanisms, to deliver highly personalized content. By analyzing

user data in real time, Netflix’s algorithms adjust recommendations based on recent behavior and contextual factors, such as the user’s device or time of day. These advanced techniques allow the system to adapt its suggestions dynamically, ensuring that the content remains contextually relevant and tailored to individual viewing habits. This adaptability is a core strength of deep learning in the streaming industry, enabling real-time, personalized engagement that evolves with the user’s preferences [4, 8, 9].

4 Evaluation of Deep Learning Approaches

Evaluating deep learning models is essential to ensure they are meeting goals for relevance, engagement and efficiency. Key performance indicators include accuracy and user satisfaction.

4.1 Accuracy of Recommendations

To assess the accuracy of recommendations, streaming platforms typically rely on Precision, Recall, and F_1 scores:

- **Precision** measures the proportion of recommended content that users engage with, providing insight into the relevance of the suggestions [10].
- **Recall** assesses the effectiveness of the system in delivering all relevant content to users, evaluating how well the algorithm captures user interests [10].
- The **F_1 Score** offers a comprehensive assessment by combining Precision and Recall into a single metric, allowing for a balanced evaluation of recommendation quality [10].

The formulas for these metrics are mathematically expressed as follows, where:

- a : Number of correctly recommended items by the recommendation system.
- b : Number of items that should have been recommended, but were missed.
- c : Number of incorrect items recommended by the system.

Metric	Formula
Precision	$\frac{a}{a+b}$
Recall	$\frac{a}{a+c}$
F_1 Score	$\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

Table 1: Formulas of Precision, Recall and F_1 Score

4.2 User Satisfaction

To evaluate user satisfaction, platforms track engagement metrics such as Retention Rate and Engagement Rate. These metrics help streaming platforms fine-tune their models, ensuring that recommendations remain relevant and meaningful over time. User satisfaction data is invaluable for updating algorithms to reflect changing viewer preferences [11].

To quantify these metrics, are used the following formulas:

- **Retention Rate** measures how well a platform retains its user base over time.
- **Engagement Rate** evaluates the level of interaction users have with content relative to its visibility.

Metric	Formula
Retention Rate	$\frac{\text{Ending Customers} - \text{New Customers}}{\text{Beginning Customers}} \times 100$
Engagement Rate	$\frac{\text{Total Engagements}}{\text{Total Impressions}} \times 100$

Table 2: Formulas of Retention Rate and Engagement Rate

5 NLP and VCA

5.1 Natural Language Processing

Natural Language Processing (NLP) interprets user-generated text, such as reviews, comments, search terms etc. Using sentiment analysis and topic modeling, NLP helps systems understand not only what users are watching, but how they feel about it. Sentiment analysis classifies text as positive, neutral or negative, while topic modeling categorizes comments by theme, allowing platforms to refine recommendations, based on emotional tone and specific interests [6, 12].

5.2 Video Content Analysis

Video Content Analysis (VCA) examines visual and thematic elements in video content, from genre to specific visual features. By analyzing patterns within video data, platforms can enhance recommendations with detailed content information, creating a refined viewing experience that aligns more closely with user preferences. For example, a viewer interested in "space" themes might be recommended other visually similar content [11, 12].

5.3 Comparison of NLP and VCA Techniques

NLP techniques effectively analyze textual inputs, such as reviews and comments, to understand user sentiment and preferences, enabling personalized recommendations. However, they rely on user-generated text and may struggle with linguistic nuances.

VCA techniques, on the other hand, analyze video data directly, identifying visual and thematic patterns, such as genre or motifs, making them ideal when textual data is unavailable. However, they require significant computational resources and do not capture user sentiment.

Overall, NLP excels in sentiment analysis, while VCA is better suited for visual content matching. The optimal solution is to combine both methods to achieve the most appropriate result.

6 Conclusion

In conclusion, the use of deep learning in video streaming services, such as Netflix, Amazon Prime Video, Disney+, etc., significantly enhances the personalized experience for users. Recommendation systems based on various approaches, such as Content-Based filtering, Collaborative filtering and Hybrid filtering, allow streaming platforms to predict user preferences more accurately, offering content that aligns with their interests. Additionally, group recommendation systems enable content selection for shared viewing, considering the preferences of multiple users at once, which is particularly useful for family or group use.

Deep learning leverages neural networks, such as CNNs and RNNs. CNNs effectively process visual features of content, helping platforms better understand genres, themes and other visual elements essential for making recommendations. RNNs, on the other hand, analyze patterns in how users' preferences evolve over time, ensuring the system adapts to these changing interests. This approach allows for more accurate and up-to-date recommendations.

The effectiveness of these models is evaluated through metrics, such as Precision, Recall, and F_1 Score, which measure how well the system meets user needs and preferences. These indicators help refine the algorithms continually. Moreover, Natural Language Processing (NLP) techniques and Video Content Analysis (VCA) add another layer of precision to recommendations, considering the emotional context expressed in user reviews, comments and visual elements in the video itself.

As a result of these advancements, streaming services not only provide more accurate recommendations, but also become more adaptive to changing user interests, improving engagement and user retention. This helps platforms maintain their competitive edge in an ever-evolving market and meet the demands of modern users for personalized content.

References

- [1] Sebastian Lubos, Alexander Felfernig, and Markus Tautschnig. “An overview of video recommender systems: state-of-the-art and research issues”. In: *Frontiers in Big Data* 6 (2023). DOI: 10.3389/fdata.2023.1281614.
- [2] Sony Peng, Sophort Siet, Sadriddinov Ilkhomjon, Dae-Young Kim, and Doo-Soon Park. “Integration of Deep Reinforcement Learning with Collaborative Filtering for Movie Recommendation Systems”. In: *Applied Sciences (Switzerland)* 14.3 (2024). DOI: 10.3390/app14031155.
- [3] Cach N. Dang, María N. Moreno-García, and Fernando De la Prieta. “Using hybrid deep learning models of sentiment analysis and item genres in recommender systems for streaming services”. In: *Electronics (Switzerland)* 10.20 (2021). DOI: 10.3390/electronics10202459.
- [4] Harald Steck, Linas Baltrunas, Ehtsham Elahi, Dawen Liang, Yves Raimond, and Justin Basilico. “Deep learning for recommender systems: A Netflix case study”. In: *AI Magazine* 42.3 (2021), pp. 7–18. DOI: 10.1609/aimag.v42i3.18140.
- [5] Yongheng Mu and Yun Wu. “Multimodal Movie Recommendation System Using Deep Learning”. In: *Mathematics* 11.4 (2023). DOI: 10.3390/math11040895.
- [6] Prof Hande, Hrushi Karlekar, Pranit Yeole, Aditya Likhar, and Himanshu Rangari. “NLP based Video Summarisation using Machine Learning”. In: *International Journal of Scientific Research in Science, Engineering and Technology* (Apr. 2023), pp. 456–461. DOI: 10.32628/IJSRSET2310265.
- [7] Laith Alzubaidi, Jinglan Zhang, Amjad J. Humaidi, Ayad Al-Dujaili, Ye Duan, Omran Al-Shamma, J. Santamaría, Mohammed A. Fadhel, Muthana Al-Amidie, and Laith Farhan. “Review of deep learning: concepts, CNN architectures, challenges, applications, future directions”. In: *Journal of Big Data* 8.1 (2021). DOI: 10.1186/s40537-021-00444-8.
- [8] Carlos A. Gomez-Urbe and Neil Hunt. “The netflix recommender system: Algorithms, business value, and innovation”. In: *ACM Transactions on Management Information Systems* 6.4 (2015). DOI: 10.1145/2843948.
- [9] Sambandam Jayalakshmi, Narayanan Ganesh, Robert Čep, and Janakiraman Senthil Murugan. “Movie Recommender Systems: Concepts, Methods, Challenges, and Future Directions”. In: *Sensors* 22.13 (2022). DOI: 10.3390/s22134904.
- [10] Zeshan Fayyaz, Mahsa Ebrahimian, Dina Nawara, Ahmed Ibrahim, and Rasha Kashef. “Recommendation systems: Algorithms, challenges, metrics, and business opportunities”. In: *Applied Sciences (Switzerland)* 10.21 (2020), pp. 1–20. DOI: 10.3390/app10217748.
- [11] Naima Souane, Malika Bourenane, and Yassine Douga. “Deep Reinforcement Learning-Based Approach for Video Streaming: Dynamic Adaptive Video Streaming over HTTP”. In: *Applied Sciences (Switzerland)* 13.21 (2023). DOI: 10.3390/app132111697.

- [12] Parul Saini, Krishan Kumar, Shamal Kashid, Ashray Saini, and Alok Negi. “Video summarization using deep learning techniques: a detailed analysis and investigation”. In: *Artificial Intelligence Review* 56.11 (2023), pp. 12347–12385. DOI: 10.1007/s10462-023-10444-0.